

Uzair Khan

Network Operations Engineer · Satellite Communications

U.S. Citizen · Public Trust · Available immediately

Manassas, VA

uk0430@icloud.com

linkedin.com/in/uk0430 · github.com/uk0430

CompTIA CySA+ (CS0-003, through 2027)

SUMMARY

Network Operations Engineer with 8+ years in satellite, enterprise and government network environments, most recently supporting the **DHS/FEMA satellite and IP network at Mount Weather. Hands-on iDirect, Hughes and Comtech experience** across GEO and MEO (O3b/SKALA) constellations — terminal restoration, beam switching, P1dB test coordination, site commissioning and real-time capacity management for maritime, aero/IFC and fixed VSAT customers. Solid Layer 2/3 and routing background (BGP, OSPF, Cisco, Juniper, Palo Alto), plus systems administration with Intune, Azure AD/Entra ID and SCCM. Strong in incident command, root-cause analysis, change management and technical documentation.

iDirect O3b MEO / SKALA DVB-S2 / ModCod P1dB Testing Beam Switching VSAT Commissioning BGP / OSPF
Cisco / Juniper Palo Alto Splunk SolarWinds Intune / Entra ID ServiceNow

PROFESSIONAL EXPERIENCE

NOC Specialist — Satellite Communications

01/2026 – 10/2026

Apex Systems, contractor to AT&T Global Public Sector (DHS/FEMA) · Mount Weather Emergency Operations Center, VA

- **NOC Operations:** Monitored, troubleshoot and configured the DHS/FEMA voice, data and satellite network on a weekend and after-hours rotation, maintaining operational readiness for a mission-critical continuity-of-government environment.
- **Satellite Hub Operations:** Operated earth station equipment at an iDirect satellite hub supporting FEMA mobile units; changed space-segment bandwidth assignments for new and changed mission requirements.
- **Troubleshooting & Restoration:** Performed fault isolation, root-cause analysis and service restoration across integrated satellite, LAN/WAN and voice systems; supported network changes and testing with engineering teams.
- **Reporting & Remote Access:** Produced bandwidth usage, traffic analysis and daily schedule reports for the government task order; maintained remote access servers and client systems, including patches and updates.

Systems Administrator / IT Functional Analyst II

12/2024 – 12/2025

MDThink – Maryland Department of IT (DoIT): Maryland Benefits (SNAP/DHS) · Linthicum, MD

- **Endpoint Fleet Management:** Managed the lifecycle of 200+ Windows 10/11 devices with Microsoft Intune, SCCM and Autopilot — imaging, role-based access, VPN and application packages for day-one onboarding.
- **Identity & Access:** Administered Azure AD/Entra ID conditional access, group-based licensing and MFA enforcement; owned Jira-based IAM requests, provisioning, de-provisioning and access reviews.
- **Policy & Patch Automation:** Wrote PowerShell scripts deployed via SolarWinds to verify GPO/Intune baseline propagation across hundreds of endpoints; ran firmware and patch deployments under change control.
- **Asset Inventory Automation:** Built a Python/PowerShell scan script to inventory state-owned and vendor devices, feeding a custom Jira schema and automated Confluence reports for the VP and management.
- **Documentation & Dashboards:** Authored KB articles and SOPs adopted organization-wide; built Jira (JQL) dashboards tracking queues, shipping and device lifecycle; prototyped an AI helpdesk chatbot for Google Spaces.
- **Physical Infrastructure:** Audited wall jacks and switch-port assignments, labeled patch panels and maintained network closets; provided on-site AV and network support for executive presentations.

Network Engineer — Network Operations Center (SATCOM)

02/2022 – 11/2024

SES Networks · Manassas, VA

24/7 Tier 2/3 NOC for a global satellite and IP network serving maritime vessels, fixed VSAT and aero (inflight connectivity) across GEO and MEO constellations. Primary NOC contact for SKALA (O3b MEO) customers and Virtual Network Operators (VNOs).

- **Terminal Restoration & Configuration:** Logged into iDirect, Hughes and Comtech terminals over in-band and out-of-band paths (management VLAN, cellular backup, Cisco ASA VPN) to clear faults, push configuration changes and restore customer traffic during service-affecting events.
- **iDirect Platform Operations:** Administered terminals via iVantage NMS and iQ Desktop — Eb/No, Ec/No, RX/TX power, ACM state, symbol rate, BER and burst timing; analyzed modem logs (network_entry, burst_timing, demod_lock) to isolate RF vs. network-layer faults.
- **Beam Switching & Capacity:** Executed manual beam switches for maritime vessels via GUI profiles and modem CLI; rebalanced vessel distribution to relieve beam congestion and managed bandwidth package changes with capacity planning.
- **SKALA / O3b MEO:** Managed SKALA bandwidth packages via GUI and CLI, monitored MEO satellite handoffs and intervened on failed handoff events for SKALA VNO customers.
- **DVB-S2 / RF Management:** Made real-time ModCod adjustments on DVB-S2 forward and return carriers; managed SCPC/MCPC carriers; distinguished rain fade from RFI and escalated spectrum analysis to RF engineering.
- **Aero Operations:** Monitored live inflight routes through airline ops dashboards, managing bandwidth allocation, beam handoffs and ModCod transitions to hold SLA throughput for cabin IFC.

- **P1dB Test Coordination:** Bridged P1dB transmit power tests between site technicians, ground station RF engineers and the NOC — stepped power increments, documented compression points, locked approved TX ceilings into NMS profiles and cleared carriers for service.
- **Site Commissioning:** Commissioned fixed and maritime VSAT terminals end-to-end — azimuth/elevation verification, modem registration, link-margin validation, P1dB test, activation and as-builts; recommissioned after equipment swaps and dry-dock returns; decommissioned legacy sites.
- **Network Layer & Monitoring:** Monitored OSPF backbone links; ran BGP diagnostics on Juniper MX5 vessel CPE (neighbor state, prefix advertisements, session resets); correlated SNMP, NetFlow and syslog data in SATNMS, Zenoss and Wireshark.
- **Splunk SIEM:** Onboarded 30+ ground station and network log sources; built SPL searches and threshold alerts for recurring fault patterns, reducing MTTD.
- **Incident & Change Management:** Led incident bridges across satellite ops, terrestrial engineering, ground stations and vendor NOCs; authored RCA reports and customer change-request notices in ServiceNow; mentored junior analysts and wrote commissioning runbooks and P1dB procedures.

Impact: Maintained 99.9%+ availability across maritime and aero links. Splunk integration improved fault correlation across 30+ nodes. P1dB bridge coordination reduced SKALA and VNO customer activation delays.

Enterprise Network Technician — Network Operations Center

01/2021 – 01/2022

ViaPath · Falls Church, VA

- **WAN/LAN Monitoring:** Monitored 100+ correctional facility sites in SolarWinds NPM/NTA; responded to SNMP alerts within 15-minute SLA windows and ran root-cause diagnostics on outages and degradation.
- **Changes & Firewall:** Performed controlled network changes and implementation testing, including Palo Alto firewall rules and routing/switching; tracked maintenance through structured CR processes.
- **Carrier Coordination & Failover:** Worked with on-site contacts and LECs to validate circuit disruptions; activated ViaSat satellite failover circuits for remote sites and documented restoration runbooks.
- **Site Lifecycle:** Provisioned new facility sites, decommissioned legacy locations and dispatched field technicians with hardware replacement SOPs.

Enterprise Account & Operations Support

07/2017 – 11/2020

Carfax · Centreville, VA

- Tier 2 support for CARFAX RESTful API and POS integrations; diagnosed data transmission, field mapping and OAuth issues between dealer management systems and vehicle history APIs.
- Remotely deployed integration software on client POS systems; managed tickets in Salesforce and escalated API defects to engineering with reproduction steps and log artifacts.

Retail Operations Manager

2012 – 2016

AT&T Mobility · Oakton, VA

- Used AT&T backend CLI provisioning to activate service, push SIM configurations and resolve carrier-level provisioning failures; activated DIRECTV satellite installs through DSAT and coordinated technician dispatch.
- Configured MDM, enterprise email, VPN and MAM policies for business accounts; led a team of 10 and owned store revenue targets.

TECHNICAL SKILLS

Satellite / RF	iDirect (iVantage, iQ Desktop), Hughes, Comtech CDM, O3b MEO/SKALA, DVB-S2, SCPC/MCPC, ModCod/ACM, P1dB testing, beam switching, VSAT commissioning, HTS/FSS, aero IFC, earth station / hub operations
Networking	BGP, OSPF, EIGRP, MPLS, VLANs, STP, LACP, TCP/IP, DNS, DHCP, NTP, VPN, LAN/WAN/MAN · Cisco ASA/IOS/NX-OS, Juniper MX/EX (JunOS), Palo Alto
Monitoring Systems	SolarWinds NPM/NTA, Splunk, SATNMS, Zenoss, SNMP, NetFlow, syslog, Wireshark, tcpdump
Automation / ITSM	Windows 10/11, Linux, Microsoft Intune, SCCM, Autopilot, Azure AD/Entra ID, Active Directory, GPO, VMware
Documentation	Python, PowerShell, bash, Splunk SPL · ServiceNow, Jira (JQL, dashboards), Confluence
	SOP/KB authoring, RCA reports, change requests, commissioning runbooks, as-builts

PROJECTS

Home Network Monitoring Lab (Raspberry Pi): Home NOC running 15 Docker services — Grafana, Prometheus, Cowrie SSH honeypot, Pi-hole + Unbound DNS (DNSSEC), fail2ban, Tailscale VPN, ntfy alerting; led a capacity-driven incident and migration plan.

TankPit Discord Bot & RAG AI System · github.com/uk0430: Hybrid RAG + SQL AI engine on local LLM inference (Mistral 7B / llama.cpp) with async scrapers feeding a 13,250-player SQLite and vector store. Python, asyncio, Docker.

EDUCATION, CERTIFICATIONS & CLEARANCE

CompTIA CySA+ (CS0-003) — valid through 2027 · **Clearance:** Public Trust · **U.S. Citizen**

Strayer University — B.S. Information Technology, expected 2028 (currently enrolled; completing prerequisite coursework at NVCC)

Northern Virginia Community College — A.A.S. Business Administration